# Exact finite buffers for the actual rational-kernel inverse

Producer proof report; unsigned PCC standing

8 November 2026

The coefficientwise inverse now has a finite modular-buffer specification connected to the actual source denominator. It no longer assumes that a caller supplies a full approximation and its exact inverse. All statements below have `[pcc:unverified]` standing pending a trusted independent signature. No new experiment or benchmark was run.

## Exact coefficient precision

Let $A = \{r \in \mathbb{Q} : |r|_2 \le 1\}$. For every commutative ring $R$, scalar $c \in R$, jet length $q$ and series $f$,

$$f \in (C(c), X^q) \quad \Longleftrightarrow \quad c \mid [X^n]f \text{ for every } n < q.$$

`FiniteBuffers.jet_mem_iff_coeff_divisible` proves both directions. For the reverse implication it constructs the coefficient witnesses, then factors the remainder by $X^q$. There is no nonzero-scalar, domain or completeness assumption.

`buffer_eq_iff_jet` specializes this to $c = 2^p$ over $A$: two length-$q$ coefficient vectors in $A/(2^p)$ agree exactly when their series differ by the actual jet ideal $(2^p, X^q)$. Claim `PCC-20261108-ff4290` pins the statement.

## Finite inverse constructor

`liftBuffer` zero-extends a vector indexed by $\mathrm{Fin}\, q$. `invertBuffer` takes that vector and its known constant unit, applies the standard inverse coefficient recurrence, and returns the first $q$ coefficients. For the actual denominator $D$, the genuine mapped source unit $\rho_m$ has the required constant value. The case $q = 0$ is the unique empty vector.

`actual_invertBuffer_correct` proves equality to the actual inverse buffer. It does not ask the caller for full series $E, F$ with $EF = 1$; the proof constructs them internally from the finite input. The general low-coefficient lemma explains why input coefficients at or above $q$ cannot affect the returned coefficients. Claim `PCC-20261108-24cbda` includes this actual-source scope.

## Concrete modular encoding

The restriction of the standard two-adic residue map gives a ring map $A \to \mathbb{Z}/2^p\mathbb{Z}$. `ResidueEncoding.residue_kernel` proves its kernel is exactly $(2^p) \subset A$. Surjectivity and the first isomorphism theorem give the actual ring equivalence

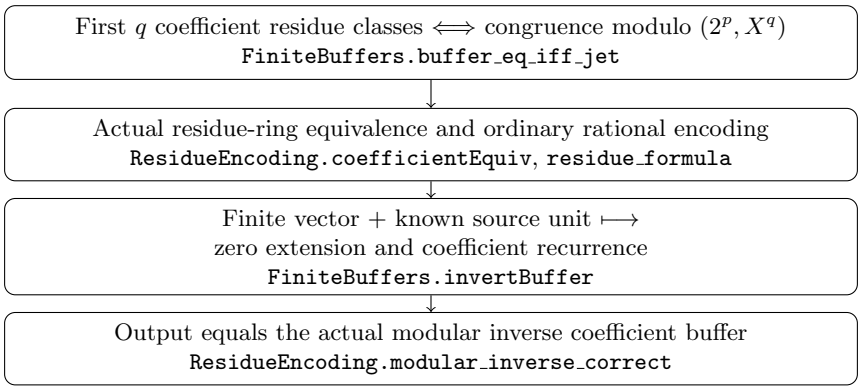
$$A/(2^p) \;\simeq\; \mathrm{ZMod}(2^p).$$

`residue_formula` identifies this map with

$$r \longmapsto \mathrm{num}(r)\,\mathrm{den}(r)^{-1} \pmod{2^p}.$$

The denominator is proved to be a unit from the actual norm bound; no p-adic approximation oracle is required to use the formula. `modular_inverse_correct` proves the finite operator correct in this modular encoding for every actual source scale/node and every $p, q$.

## Proof chain and representation scope




`modular_buffer_card` proves exactly $2^{pq}$ possible length-$q$ modular vectors. This is an information/cardinality statement, not a measurement of a program's allocation. Claim `PCC-20261108-ffec83` records the ring equivalence, residue formula and cardinality.

The current Lean operator uses mathematical structures marked noncomputable. The residue formula supplies its ordinary finite arithmetic interpretation, but these proofs do not certify a compiler, a native implementation or a bit-operation bound. Such claims require a separate refinement and cost proof.

## Verification and provenance

The mathematical source is in `lean/FiniteBuffers.lean` and `lean/ResidueEncoding.lean`; source commit `07cfb5586` contains them. The default Lake build includes both modules. The actual expanded source gate checks transitive axioms; the official import-inclusive fresh replay and status receipts are recorded in `lean/FINITE_BUFFERS_VERIFICATION.md`. All failed elaborations remain in the evidence directory, not accepted as proofs. No checker, kernel or trust list was changed.

The original empirical protocol, instrument, raw data, export, plots and PDFs remain unchanged. This is a new versioned proof report. No additional panel, timing run or cloud job occurred. The current main merge preserves both appended session logs; the PCC main-prefix check passes without rewriting events.

CI caught missing declaration hashes in unsigned claim metadata. The original records are archived; canonical pins and corrected registrations were appended. No theorem, verifier or trust policy changed. See `pcc-pin-repair/`.

Sources checked: the pinned Mathlib power-series coefficient and $X$-divisibility APIs, ideal-span/quotient APIs, PadicInt residue maps and kernel, rational denominator unit, and ZMod inverse/cardinality APIs. KB queries were unavailable (status127); primary-source fallback, no priority or literature-exhaustion claim.

## The remaining asymptotic obligation

The local finite representation is now explicit. The outstanding obligations are actual contraction homogeneity, full finite-precision Hermite recovery, and uniform all-level modes with construction and child-call bit costs. These local algebraic statements neither supply nor bound that complete recursive algorithm. See `PLAN.md`; no sub-exponential algorithm theorem or unrestricted nonexistence theorem is asserted by this report.